
ISA 632 – Big Data Analytics & Modern AI

Group Project

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Build a cloud-based GenAI Application

- Use Databricks services to create an GenAI chatbot for a business scenario that you select.
- Tasks
 - Analyze the proposed use cases' potential for success
 - Collect data as context information
 - Practice the key GenAI technologies
 - Create a prototype on Databricks to demonstrate the feasibility
 - Share your findings and experiences with a group of GenAI experts

People's Choice Award in Previous Cohorts

BagelBot - Find your Fantastic Bagels



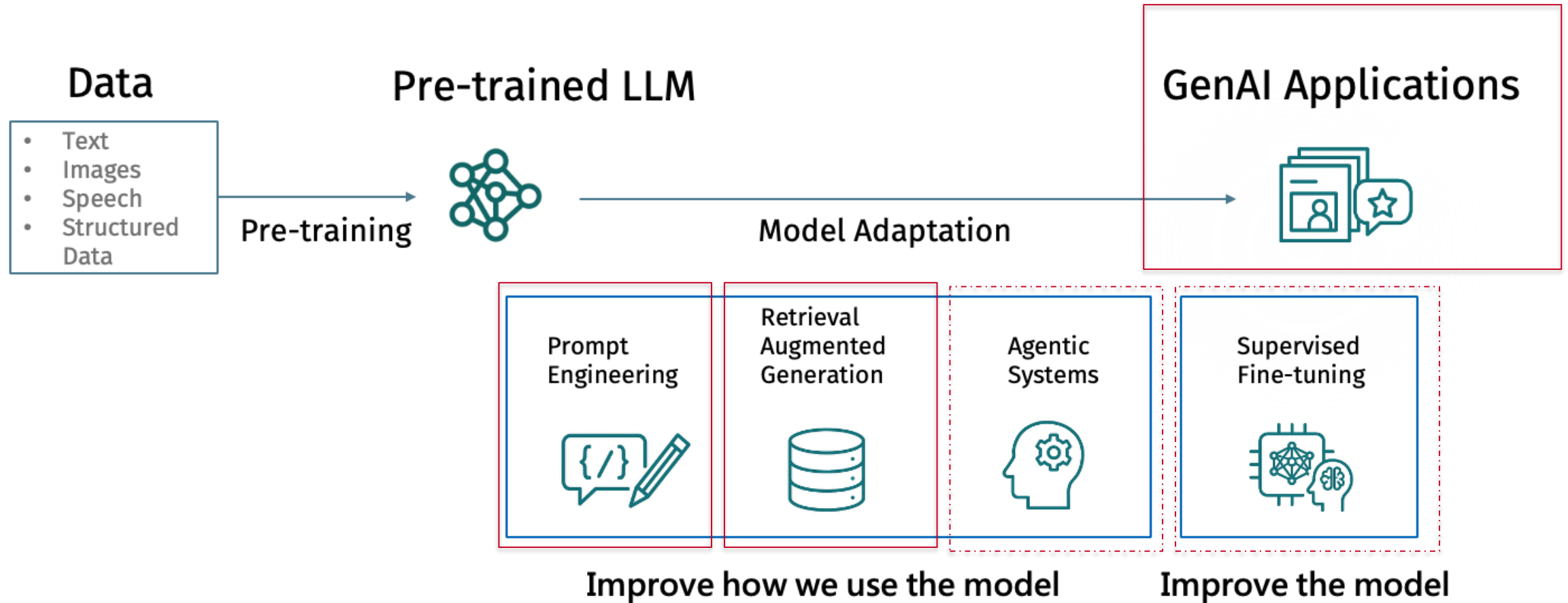
CLAIRE LINN

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HAYDEN FOGLE

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Prototype Architecture



Prototype Components

➤ 4 Major Components

- System Prompt
- RAG
- Evaluation
- Deployment

➤ Optional

- Agentic AI
- LLM Fine-tuning
- Streamlit App
- ...

Potential Data Sources

- Collected data in your ISA514 group project
- Databricks marketplace
 - Filter by Free and Files/Tables, Search term: text
 - E.g., Databricks Documentation Dataset
 - <https://dbc-0726d26f-3749.cloud.databricks.com/marketplace/consumer/listings/03bbb5c0-983d-4523-833a-57e994d76b3b?o=7474656346303369>
- Other public data sources
 - Yelp Open Dataset
 - <https://business.yelp.com/data/resources/open-dataset/>
- Tips
 - Large-scale data collection effort is **NOT** recommended.
 - Start with a small sample (1000 rows) to build RAG before running on the full one

Sample Business GenAI Chatbot

➤ FSB Advising Chatbot

- https://www.fsb.miamioh.edu/fsb/advising_chat/

➤ SIGHT Project

- <https://sight.fsb.miamioh.edu/>

Presentation

- Each team will present to a group of GenAI experts.
- Arrive before the assigned time slot
- Each team will have 10 minutes for presentation, including preparation and Q&A.
- Let one member take charge of the slides and have a backup too.

Guidelines

- Think that you are a team of consultants presenting to a client. Make a professional presentation with impressive demo of your project.
- The presentation should show the major contributions of your project. Also, make a few slides to briefly discuss your motivation, background, study design & methodology, challenges faced and where to take the project further from here. The design of the presentation is totally up to your team.
- Every member should participate the presentation, including presenting slides, demoing tools and results, answering questions, etc. It should not be solo.

Tips

- It will be very helpful if you do several practice rehearsals beforehand. Time management is the key.
- Don't include many slides in main presentation. Put them as backup slides.
- Make good use of the Q&A time to further explain some technical details (e.g., use those backup slides).
- If real-time demo is risky, video-tape it in advance. Zoom is a convenient tool for video recording.
- If the video file is too large to carry, upload it to YouTube, and insert a link in your slides.

Final Deliverables

- Each team should submit their slides after their final presentation.
- A complete website/app with runnable chatbot
- A brief technical document includes a short introduction about each major technical component in your project, such as data collection, system prompt, vector store, etc. So other people can follow your workflow and replicate the app.
- Companion files include all developed notebooks, scripts, and data used in your project.

Rubrics

➤ Business Value & Problem Framing

- Clarity of business use case
- Relevance and realism of the scenario
- Justification of why GenAI is needed (vs. traditional solutions)
- Potential impact / value

➤ Data & Context Engineering

- Quality and relevance of collected data
- Data preprocessing and organization
- Appropriateness for RAG / context injection

➤ Evaluation & Validation

- Defined evaluation criteria
- Testing with realistic queries
- Evidence of performance (examples, metrics, comparisons)
- Reflection on limitations

➤ Implementation & Prototype

- Functional Databricks-based prototype
- Deployment quality (e.g., UI, Streamlit, API)

➤ Presentation & Demo

- Clear, professional, consultant-style presentation
- Effective storytelling (problem → solution → impact)
- Time management (10 min constraint)

People's Choice Award

